Collecting Data from Experiments

Jonathan R. Bacolod

Experiment

An **experiment** is an activity or process performed to observe outcomes.

Example: tossing a coin.

Outcome

An **outcome** is a possible result of an experiment.

A coin toss may result in heads (H) or tails (T).

Trial

A **trial** is one performance of an experiment.

Tossing a coin once counts as one trial.

Tossing a coin 10 times is 10 trials.

Data

Data is information collected by observing and recording the outcomes of an experiment.

Example: 10 tosses of a coin

$$H,\ T,\ H,\ H,\ T,\ H,\ T,\ T,\ H,\ H$$

Frequency

The **frequency** is the number of times a particular outcome occurs during an experiment.

In the data above, H appears 6 times.

Frequency of $H = 6$

Tally Marks

Tally marks are marks used to count and organize outcomes efficiently.

Every group of five has four vertical marks crossed by a fifth mark.

5 is written as 

Tally Marks Mnemonic

Four bars, then a gate: that makes five!

Number	Tally
1	
3	
5	
7	

Frequency Table

A **frequency table** is a table that organizes outcomes and their corresponding frequencies.

Outcome	Tally	Frequency
H	$\cancel{\|\|\|\|}\,\|$	6
T	$\|\|\|\|$	4

Key Terms Summary

Term	**Meaning**
Experiment	activity to observe
Outcome	possible result
Trial	one performance
Frequency	how many times

How to Collect Data

1. Identify the experiment and the outcomes to observe.
2. Determine the number of trials to perform.
3. Perform the experiment repeatedly.

How to Organize Data

4. Record each outcome using tally marks or a data list.
5. Count the occurrences of each outcome.
6. Organize the results in a frequency table.

Check Your Work

Check that the sum of all frequencies equals the total number of trials.

Sum of frequencies $=$ Number of trials

Example: $6 + 4 = 10$

Examples

Example 1

A coin is tossed 10 times.

$H,\ T,\ H,\ H,\ T,\ H,\ T,\ T,\ H,\ H$

Make a frequency table.

Example 1 Solution

Outcome	**Tally**	**Frequency**
H	$\cancel{\|\|\|\|}\,\|$	6
T	$\|\|\|\|$	4
Total		10

Example 2

A coin is tossed 20 times. The frequency of heads is 12.
Find the frequency of tails.

$$20 - 12 = 8$$

The frequency of tails is 8.

Example 2 Solution

Outcome	**Frequency**
H	12
T	8
Total	20

Practice Exercises

Practice 1

Perform the indicated tasks.

A student tosses a coin 12 times and records the following outcomes:

$$H,\ T,\ H,\ H,\ T,\ H,\ T,\ H,\ H,\ T,\ H,\ T$$

Make a frequency table showing the number of heads and tails.

Practice 2

Perform the indicated tasks.

A student rolls a die 15 times and records these results:

$$2,\ 4,\ 1,\ 6,\ 3,\ 2,\ 5,\ 4,\ 2,\ 1,\ 3,\ 6,\ 5,\ 2,\ 4$$

Construct a frequency table for the outcomes $1, 2, 3, 4, 5,$ and 6.

Practice 3

Perform the indicated tasks.

A die is rolled 18 times. The results are classified as **Prime** or **Not Prime**. If prime numbers occur 8 times, determine the frequency of not-prime outcomes. Present the results in a frequency table.

Practice 4

Perform the indicated tasks.

A student spins a spinner 16 times. The spinner has three possible outcomes: Red, Blue, and Yellow. The frequencies of Red and Blue are 6 and 5, respectively. Determine the frequency of Yellow and make a frequency table.

Activities

Activity 1

Perform the indicated tasks.

Toss a coin 14 times. Record the outcomes in order:

$T,\ H,\ H,\ T,\ H,\ T,\ H,\ H,\ T,\ T,\ H,\ H,\ T,$

Make a frequency table showing the number of heads and tails.

Activity 2

Perform the indicated tasks.

Roll a die 16 times and record these results:

$3,\ 1,\ 5,\ 2,\ 6,\ 3,\ 4,\ 2,\ 1,\ 5,\ 6,\ 3,\ 2,\ 4,\ 5,\ 1$

Construct a frequency table for the outcomes $1, 2, 3, 4, 5,$ and 6.

Activity 3

Perform the indicated tasks.

Toss a coin 22 times. Suppose your recorded frequency of heads is 13. Determine the frequency of tails and organize the information in a frequency table.

Activity 4

Perform the indicated tasks.

Toss a coin in four separate sets of 10 tosses each. Suppose the recorded frequencies of heads are $5, 4, 8,$ and 6. Make a table showing the set number and the number of heads obtained in each set.

Thank you!